6.630 Solution to Problem Set 1

Solution P1.1

(a) $\overline{E}_1$ and $\overline{E}_3$ satisfy the wave equation with $k = \omega(\mu_0\epsilon_0)^{1/2}$,

(b) By Faraday's law

$$\frac{\partial}{\partial t}\overline{H} = -\frac{1}{\mu_0}\nabla \times \overline{E}$$

$$\overline{H}_1 = 0$$

$$\overline{H}_2 = \begin{cases} \hat{y}\dfrac{\sqrt{2}k}{\omega\mu_0}\cos\left[\omega t + k(x-z)/\sqrt{2}\right] & , \quad x - z > 0 \\ \hat{y}\dfrac{-\sqrt{2}k}{\omega\mu_0}\cos\left[\omega t + k(z-x)/\sqrt{2}\right] & , \quad x - z < 0 \end{cases}$$

$$\overline{H}_3 = (\hat{x} - \hat{z})\frac{-k}{\omega\mu_0}\cos(\omega t + ky)$$

(c) $\overline{E}_3$ qualifies as electromagnetic waves.

$\overline{E}_1$ violates Gauss' Theorem $\nabla \cdot \overline{E} = 0$.

The derivative of $\overline{E}_2$ and $\overline{H}_2$ do not exit at $x = z$, so they violate all of the Maxwell equations.

The $\overline{k}$-vector for $\overline{E}_3$ is $\overline{k}_3 = -\hat{y}k$. $\overline{E}_3$ and $\overline{H}_3$ are all perpendicular to the direction of propagation as $\overline{k}_3 \cdot \overline{E}_3 = 0$ and $\overline{k}_3 \cdot \overline{H}_3 = 0$.

Solution P1.2

(a-i) 60 Hz: $\lambda = c/f = 5 \times 10^6(m)$
(a-ii) AM radio (535–1605 kHz): $\lambda = 186.9 \sim 560.8(m)$
(a-iii) FM radio (88–108 MHz): $\lambda = 2.778 \sim 3.409(m)$
(a-iv) C-band (4–6 GHz): $\lambda = 0.05 \sim 0.075(m)$
(a-v) Visible light ($\sim 10^{14}$ Hz): $\lambda = \sim 3 \times 10^{-6}(m)$
(a-vi) X-rays ($\sim 10^{18}$ Hz): $\lambda = \sim 3 \times 10^{-10}(m)$

(b-i) 1 km: $f = c/\lambda = 3 \times 10^5(Hz)$
(b-ii) 1 m: $f = 3 \times 10^8(Hz)$
(b-iii) 1 mm: $f = 3 \times 10^{11}(Hz)$
(b-iv) 1 μm: $f = 3 \times 10^{14}(Hz)$
(b-v) 1 Å: $f = 3 \times 10^{18}(Hz)$

(c-i) 1 km: $k = 2\pi/\lambda = K_o/\lambda = 10^{-3}K_o$
(c-ii) 1 m: $k = 1K_o$
(c-iii) 1 mm: $k = 10^3K_o$
(c-iv) 1 μm: $k = 10^6K_o$
(c-v) 1 Å: $k = 10^{10}K_o$

Solution P1.3

$$\begin{aligned}
\nabla \times \left(\nabla \times \overline{E}\right) &= \nabla \times \begin{bmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ E_x & E_y & E_z \end{bmatrix} \\
&= \begin{bmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z}\right) & \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x}\right) & \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y}\right) \end{bmatrix} \\
&= \left[\frac{\partial}{\partial x}\left(\frac{\partial E_y}{\partial y} + \frac{\partial E_x}{\partial x} + \frac{\partial E_z}{\partial z}\right) - \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) E_x\right] \hat{x} \\
&+ \left[\frac{\partial}{\partial y}\left(\frac{\partial E_y}{\partial y} + \frac{\partial E_x}{\partial y} + \frac{\partial E_z}{\partial z}\right) - \left(\frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) E_y\right] \hat{y} \\
&+ \left[\frac{\partial}{\partial z}\left(\frac{\partial E_y}{\partial y} + \frac{\partial E_x}{\partial y} + \frac{\partial E_z}{\partial z}\right) - \left(\frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) E_z\right] \hat{z} \\
&= \nabla\left(\nabla \cdot \overline{E}\right) - \nabla^2 \overline{E}
\end{aligned}$$

$$\begin{aligned}
\nabla \cdot \left(\overline{E} \times \overline{H}\right) &= \nabla \cdot \begin{bmatrix} \hat{x} & \hat{y} & \hat{z} \\ E_x & E_y & E_z \\ H_x & H_y & H_z \end{bmatrix} \\
&= \frac{\partial}{\partial x}\left(E_y H_z - E_z H_y\right) + \frac{\partial}{\partial y}\left(E_z H_x - E_x H_z\right) + \frac{\partial}{\partial z}\left(E_x H_y - E_y H_x\right) \\
&= H_x\left(\frac{\partial}{\partial y} E_z - \frac{\partial}{\partial z} E_y\right) + H_z\left(\frac{\partial}{\partial x} E_y - \frac{\partial}{\partial y} E_x\right) + H_y\left(\frac{\partial}{\partial z} E_x - \frac{\partial}{\partial x} E_z\right) \\
&- E_x\left(\frac{\partial}{\partial y} H_z - \frac{\partial}{\partial z} H_y\right) - E_y\left(\frac{\partial}{\partial z} H_x - \frac{\partial}{\partial x} H_z\right) - E_z\left(\frac{\partial}{\partial x} H_y - \frac{\partial}{\partial y} H_x\right) \\
&= \overline{H} \cdot \left(\nabla \times \overline{E}\right) - \overline{E} \cdot \left(\nabla \times \overline{H}\right)
\end{aligned}$$

$$\begin{aligned}
\nabla \cdot \left(\nabla \times \overline{A}\right) &= \nabla \cdot \begin{bmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A_x & A_y & A_z \end{bmatrix} \\
&= \frac{\partial}{\partial x}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \frac{\partial}{\partial y}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \frac{\partial}{\partial z}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right) \\
&= 0
\end{aligned}$$

$$\begin{aligned}
\nabla \times (\nabla \phi) &= \begin{bmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \partial\phi/\partial x & \partial\phi/\partial y & \partial\phi/\partial z \end{bmatrix} \\
&= \left(\frac{\partial^2 \phi}{\partial y \partial z} - \frac{\partial^2 \phi}{\partial y \partial z}\right) \hat{x} + \left(\frac{\partial^2 \phi}{\partial x \partial z} - \frac{\partial^2 \phi}{\partial x \partial z}\right) \hat{y} + \left(\frac{\partial^2 \phi}{\partial x \partial y} - \frac{\partial^2 \phi}{\partial x \partial y}\right) \hat{z} \\
&= 0
\end{aligned}$$

Solution P1.4

The phase ϕ is a function of both space and time. To get the wave velocity, one needs to fix a phase and trace this constant phase in space and time. In other words, $\Delta\phi = 0$ in both space and time coordinates. Therefore, we can take derivative with respect to time or space to get the wave velocity.

If k is increased, the slope tends to be flat, meaning the wave velocity decreased. In addition, we can see that the wave becomes denser in space.

6.630 Solution to Problem Set 2

Solution P2.1

AM (i): Vertically oriented.
TV (ii): The antenna is parallel to the ground and perpendicular to the direction from which the signal comes.
FM (i)(ii): The antenna may be oriented in any direction as long as the antenna is on the plane perpendicular to the direction from which the signal is coming. This is because the antenna will always intercept the electric field that is rotating around a circular path in that plane.

Solution P2.2

(a) $6e^{i\frac{\pi}{4}}$
(b) $8e^{i\frac{\pi}{2}}$
(c) $-3i-2$
(d) $-6i$
(e) None
(f) None

Solution P2.3

(a)

$$\overline{E} = \hat{x}2\cos(kz-\omega t+\frac{\pi}{2}) + \hat{y}\cos(kz-\omega t+\frac{\pi}{4})$$

At the point $z=z_0$, $\overline{E}$ will trace a trajectory

$$x = 2\cos(kz_0-\omega t+\frac{\pi}{2}) = -2\sin(kz_0-\omega t)$$
$$y = \cos(kz-\omega t+\frac{\pi}{4}) = \frac{1}{\sqrt{2}}\cos(kz_0-\omega t) - \frac{1}{\sqrt{2}}\sin(kz_0-\omega t)$$

The equation describing this trajectory is

$$\frac{x^2}{2} - \sqrt{2}xy + 2y^2 = 1$$

which is the equation of an ellipse. Therefore, the wave is elliptically polarized.

(b) Consider the following linearly polarized plane wave:

$$\begin{aligned}\overline{E} &= \hat{x}\cos(kz-\omega t)\\ &= \left[\frac{1}{2}\hat{x}\cos(kx-\omega t) + \frac{1}{2}\hat{y}\sin(kz-\omega t)\right] + \left[\frac{1}{2}\hat{x}\cos(kx-\omega t) - \frac{1}{2}\hat{y}\sin(kz-\omega t)\right]\end{aligned}$$

This is the superposition of two circularly polarized waves.

(c) Consider the following circularly polarized plane wave:

$$\overline{E} = \hat{x}\cos(kz-\omega t+\pi/4) + \hat{y}\cos(kz-\omega t-\pi/4)$$

This is the superposition of two linearly polarized waves.

Solution P2.4

$\lambda = \frac{1}{A} = \frac{1}{100} = 0.01\,(\mathrm{m})$

The wave propagates in the $+\hat{z}$ direction.
At $z = 0$,

$$\overline{E}(z=0,t) = E_0\left[\hat{x}\cos(\omega t) - \hat{y}\sin(\omega t)\right]$$

This is a left-hand circularly polarized plane wave.
At $t = 0$,

$$E_x = E_0\cos\left(\frac{2\pi}{\lambda}z\right)$$

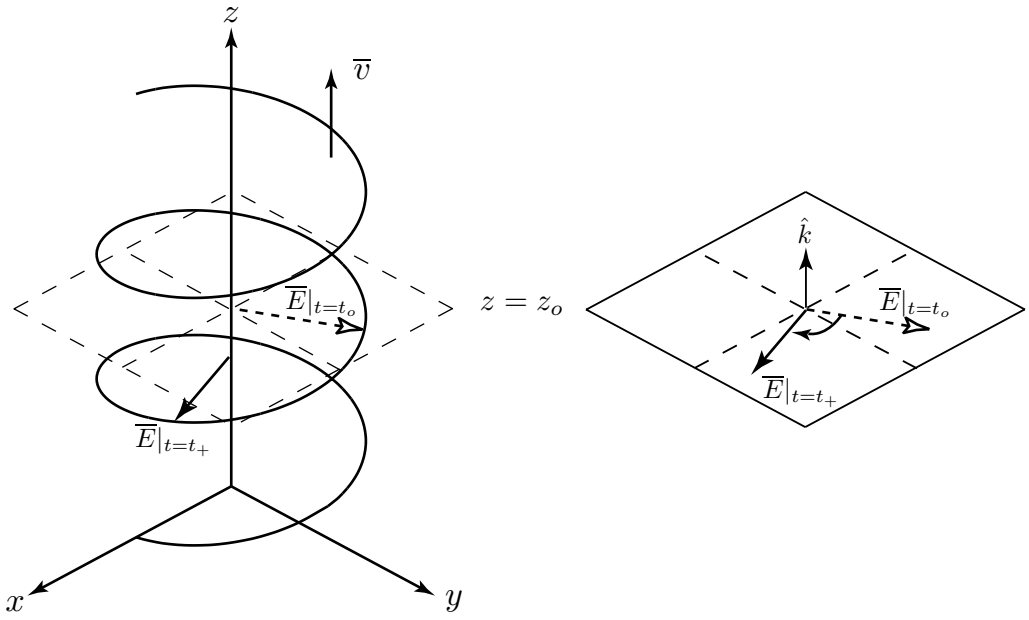
$$E_y = E_0\sin\left(\frac{2\pi}{\lambda}z\right)$$

The parametric equation of a helix is

$$x = R\cos\left(\frac{2\pi}{p}z\right)$$

$$y = R\sin\left(\frac{2\pi}{p}z\right)$$

where p is the pitch of the helix. Thus the locus of the tip point of the electric field vector measured along the z axis is a right-handed helix with the pitch $p = \lambda$. From the figure below, we see that $\overline{E}|t_0 \times \overline{E}|t_+$ is in the direction of $-\hat{k}$, thus the wave is temporally l.h.c.p.



At $\omega t = \pi/4$,

$$E_x = E_0\cos\left[\frac{2\pi}{\lambda}(z - \frac{\lambda}{8})\right]$$

$$E_y = E_0\sin\left[\frac{2\pi}{\lambda}(z - \frac{\lambda}{8})\right]$$

The helical locus advances along $+\hat{z}$ without rotating!

Solution P2.5

(a) Speed of light $c = 3 \times 10^8$m/sec
Distance between Sun and Earth $D = 150 \times 10^9$m
Travelling time of light from Sun to Earth $T = D/C = 500\text{sec} = 8.33\text{min}$

(b) Power received by the Earth $P_r = S(\text{power density}) \times A(\text{cross section area})$

$$A = \pi r^2 = \pi \times (6.4 \times 10^6)^2 \text{m}^2$$
$$S = 1.5\text{kW/m}^2$$
$$P_r = 1.93 \times 10^{14}\text{kW}$$

(c) The total power radiated by the Sun is

$$P_{total} = S(\text{power density at distance } D) \times A_T(\text{Total surface area of sphere with radius } D)$$
$$A_T = 4\pi D^2 = 4\pi \times (150 \times 10^9)^2 \text{m}^2$$
$$S = 1.5\text{kW/m}^2$$
$$\text{Mass of Sun } m = 2 \times 10^{30}\text{kg}$$
$$\text{Efficiency } E_{eff} = 1\%$$

Total time that Sun can radiate

$$T = \frac{mc^2 E_{eff}}{P_{Total}} = \frac{mc^2 E_{eff}}{S 4\pi D^2}$$
$$= \frac{(2 \times 10^{30}) \times (3 \times 10^8)^2 \times 0.01}{(1.5 \times 10^3) \times 4\pi \times (150 \times 10^9)^2} = 4.2 \times 10^{18}\text{sec} \approx 1.34 \times 10^{11}\text{years}$$

(d)

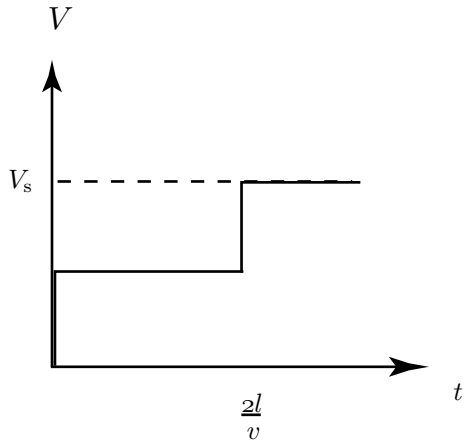
$$\text{Poynting power } S = P(\text{power density per Hz}) \times W(\text{bandwith})$$
$$= 10^{-20}\text{Wm}^{-2}\text{Hz}^{-1} \times 10^9\text{Hz} = 10^{-11}\text{Wm}^{-2}$$
$$\text{Since } S = E^2/2\eta, \text{ we have}$$
$$E = \sqrt{2\eta S} = \sqrt{2 \times 120\pi \times 10^{-11}} = 8.68 \times 10^{-5} \frac{\text{volt}}{\text{m}}$$

6.630 Solution to Problem Set 5

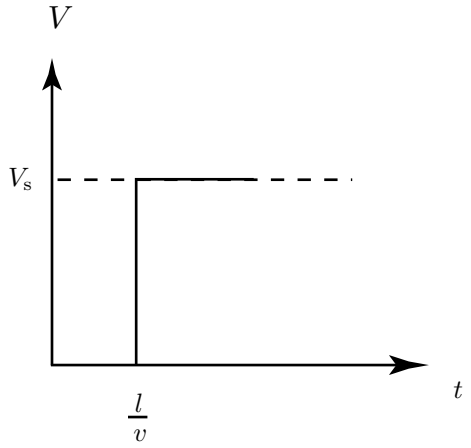
Solution P5.1

(a) $t = 2l/v = 4$ ns

(b) The signal quality at the source is poor with two separate transitions to arrive at V_s.



(c) The signal quality at the load is good with a single transition to V_s.



Solution P5.2

(a) $V_+ = I_o Z_o/2, V_- = -I_o Z_o/2$.

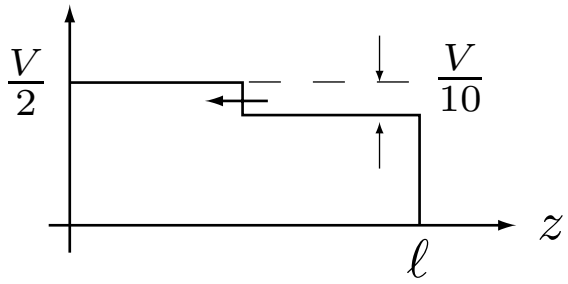
(b) $\Gamma_L = V_-/V_+ = -1$.

(c) $Z_L = Z_o \frac{1+\Gamma_L}{1-\Gamma_L} = 0$.

(d) At $\omega t = \pi/2$, $V(z,t) = \mathrm{Re}\{V(z)e^{j\omega t}\} = I_o Z_o \sin kz$.
(e) $Z_s = V_s/I_{in} = Z_o$.

Solution P5.3

(a) The load of $T1$ is Z_o parallel with $2Z_o$, which is $\frac{2}{3}Z_o$. Therefore the reflection coefficient $\Gamma_L = -\frac{1}{5}$.



(b) The input impedance of $T3$ is zero since its length is $\lambda/2$ and its load is short. Therefore the input impedance of $T2$ parralel with $T3$ is zero. In orther words, the load of $T1$ is short. Therefore Z_{in} for $T1$ is zero.

6.630 Solution to Problem Set 3

Solution P3.1

$mv^2/R = Ze^2/4\pi\epsilon R^2$ gives $R = 4\pi\epsilon n^2\hbar^2/Zme^2 \approx 0.52\,n^2 \times 10^{-10}\,\mathrm{m}$ for $Z = 1$.

Solution P3.2

The wave number for the attenuation term is

$$k_I = \omega\sqrt{\mu_o\epsilon}\left[\frac{1}{2}\left(\sqrt{1+\frac{\sigma^2}{\epsilon^2\omega^2}}-1\right)\right]^{1/2}.$$

For a good conductor,

$$\frac{\sigma^2}{\epsilon^2\omega^2} \gg 1,$$

we have

$$k_I \approx \left(\frac{\sigma\mu_o\omega}{2}\right)^{1/2},$$

thus

$$d_p = \frac{1}{k_I} \approx \sqrt{\frac{2}{\omega\mu_o\sigma}}.$$

For the material with low conductivity,

$$\frac{\sigma^2}{\epsilon^2\omega^2} \ll 1,$$

we can approximate

$$\sqrt{1+\frac{\sigma^2}{\epsilon^2\omega^2}} \approx 1+\frac{\sigma^2}{2\epsilon^2\omega^2}.$$

Thus

$$k_I \approx \frac{\sigma\eta}{2},$$

and

$$d_p = \frac{1}{k_I} \approx \frac{2}{\sigma\eta}$$

(a) Since the term,

$$\frac{\sigma}{\epsilon\omega} = \frac{2}{40\times 8.85\times 10^{-12}\times 2\pi\times 2.5\times 10^9} = 0.36,$$

we get

$$\begin{aligned} k_I &= \omega\sqrt{\mu_o\epsilon}\left[\frac{1}{2}\left(\sqrt{1+\frac{\sigma^2}{\epsilon^2\omega^2}}-1\right)\right]^{1/2} \\ &= \frac{2\pi\times 2.5\times 10^9}{3\times 10^8}\sqrt{40}\left[\frac{1}{2}\left(\sqrt{1+0.36^2}-1\right)\right]^{1/2} = 58.7(m^{-1}), \end{aligned}$$

thus

$$d_p = \frac{1}{k_I} = \frac{1}{58.7}(m) = 1.7(cm).$$

For polystyrene foam,

$$\frac{\sigma}{\epsilon\omega} = \frac{4 \times 10^{-6}}{1.03 \times 8.85 \times 10^{-12} \times 2\pi \times 2.5 \times 10^{9}} = 2.79 \times 10^{-5} \ll 1,$$

we get

$$d_p \approx \frac{2}{\sigma\eta} = \frac{2\sqrt{1.03}}{4 \times 10^{-6} \times 377} = 1346.0(m)$$

(b) For good conductor,

$$\frac{\epsilon\omega}{\sigma} < 0.1$$

thus

$$\omega < 0.1 \times \frac{\sigma}{\epsilon} = 0.1 \times \frac{5 \times 10^{-3}}{10 \times 8.85 \times 10^{-12}} = 5.65(rad/sec),$$

or

$$f_{\max} = \frac{\omega}{2\pi} = 0.899(MHz)$$

(c) Since $\frac{\sigma}{\epsilon\omega} \gg 1$, thus

$$d_p \approx \sqrt{\frac{2}{\omega\mu_o\sigma}} = \sqrt{\frac{2}{2\pi \times 10^8 \times 4\pi \times 10^{-7} \times 3.54 \times 10^7}} = 8.46 \times 10^{-6}(m).$$

So the thickness of the aluminum layer is $5d_p = 4.24 \times 10^{-5}(m)$. The thickness of ordinary aluminum foil is

$$10^{-3}(inch) = 2.54 \times 10^{-5}(m) < 4.24 \times 10^{-5}(m),$$

which is not thick enough.

(d)

(1) For $f = 100(Hz)$,

$$\frac{\sigma}{\omega\epsilon} = \frac{4}{2\pi \times 100 \times 80 \times 8.85 \times 10^{-12}} = 8.99 \times 10^{6},$$

and skin depth

$$d_p \approx \sqrt{\frac{2}{\omega\mu_o\sigma}} = \sqrt{\frac{2}{2\pi \times 100 \times 4\pi \times 10^{-7} \times 4}} = 25.2(m).$$

(2) For $f = 5(MHz)$, the loss tangent is

$$\frac{\sigma}{\omega\epsilon} = \frac{4}{2\pi \times 5 \times 10^6 \times 80 \times 8.85 \times 10^{-12}} = 180,$$

and the skin depth is

$$d_p \approx \sqrt{\frac{2}{\omega\mu_o\sigma}} = \sqrt{\frac{2}{2\pi \times 5 \times 10^6 \times 4\pi \times 10^{-7} \times 4}} = 0.11(m).$$

(e) The penetration depth at frequency $f = 1.0(kHz)$ is

$$d_p \approx \sqrt{\frac{2}{\omega\mu_o\sigma}} = \sqrt{\frac{2}{2\pi \times 1000 \times 4\pi \times 10^{-7} \times 4}} = 7.96(m) = \frac{1}{k_I}.$$

The attenuation term for the radiation power is

$$e^{-2k_I z} = e^{-2\times\frac{100}{7.96}} = 1.22\times10^{-11} = -109.1(dB).$$

Solution P3.3

(a)

$$\overline{T} = \hat{r}\times\overline{F} = \frac{1}{2}l_b\hat{x}\times(\hat{z}\times\hat{x}Il_zB_0) - \frac{1}{2}l_b\hat{x}\times(-\hat{z}\times\hat{x}Il_zB_0) = \hat{z}IAB_0$$

From $\overline{M} = -\hat{y}IA$, and $\overline{B} = \hat{x}B_0$, we find that, $\overline{T} = \overline{M}\times\overline{B} = \hat{z}IAB_0$.

(b) The magnetic moment, $\overline{M}$ of the current loop can be found by applying $\overline{M} = \hat{n}A_0I_l = \frac{1}{\sqrt{2}}(\hat{x}-\hat{y})A_0I_l$.

To find the magnetic field, $\overline{H}$, at the position of the loop due to the straight wire carrying current, I_0, we use the integral form of Ampere's Law, that is,

$$\oint_C \overline{H}\cdot d\bar{l} = \int_s \overline{J}\cdot ds = I_0$$

By symmetry the $\overline{H}$ field will be of uniform amplitude at a fixed radius and in the $\hat{\phi}$ direction yielding,

$$\oint_C \overline{H}\cdot d\bar{l} = \int_0^{2\pi} H_\phi d\ d\phi = 2\pi dH_\phi$$

which implies that at the loop's postion

$$\overline{B} = \hat{\phi}\frac{I_0\mu_0}{2\pi d} = \hat{x}\frac{I_0\mu_0}{2\pi d}$$

where the fact that $\overline{B} = \mu_0\overline{H}$ was used. To calculate the torque, we apply,

$$\overline{T} = \overline{M}\times\overline{B} = \hat{z}\frac{A_0I_lI_0\mu_0}{2\sqrt{2}\pi d}$$

which means that the current loop will move about the z-axis in a counter-clockwise direction.

6.632 Solution to Problem Set 2

Solution P2.1

(a) The penetration depth is 2 cm.
(b) At 60 Hz, loss tangent is 1.5×10^7 and penetration depth 32 m.
At 10 MHz, loss tangent is 90 and penetration depth 8 cm.
(c) $|\overline{E}|$ is 1 volt/m just beneath surface and 0.019 volt/m at 100 m depth. Power density is $25\,\text{watt/m}^2$ just beneath surface and $8.39\,\text{mW/m}^2$ at 100 m depth.

Solution P2.2

(b) $N = 7 \times 10^{28}\,\text{m}^{-3}$

$$d_p = \sqrt{\frac{9.1 \times 10^{-31}}{7 \times 10^{28} \times (1.6 \times 10^{-19})^2 \times 4\pi \times 10^{-7}}} \simeq 2 \times 10^{-8}\,\text{m}$$

(c) The skin depth of a good conductor $\delta \propto 1/\sqrt{f}$. When the frequency is very low, δ becomes large; hence a field can penetrate it. For superconductors, however, the penetration depth is independent of frequency. This suggests that even slowly-varying fields (E and H) cannot penetrate the superconductor. Note that a steady current can exist within the superconductor without the support of an electric field.

Solution P2.3

When the wave frequency is much greater than the electron collision frequency, the collision effect can be neglected and the plasma can be treated as a lossless medium. Using Lorentz force law and Newton's law, we have

$$m\frac{d\overline{v}}{dt} = q(\overline{E} + \overline{v} \times \overline{B})$$

Under the time-harmonic excitation and with $\overline{v}$ much smaller than c, we find

$$-i\omega m\overline{v} = q(\overline{E} + \overline{v} \times \overline{B}_0)$$

Defining a vector $\overline{\omega}_c = q\overline{B}_0/m$, we write

$$\overline{\omega}_c \times \overline{v} = \frac{q}{m}\overline{E} + i\omega\overline{v}$$

we cross-multiply and dot-multiply the above equation by $\overline{\omega}_c$.

$$-i\omega\overline{\omega}_c \times \overline{v} = \frac{q}{m}\overline{\omega}_c \times \overline{E} + \omega_c^2\overline{v} - \overline{\omega}_c(\overline{\omega}_c \cdot \overline{v})$$
$$-i\omega\overline{\omega}_c \cdot \overline{v} = \frac{q}{m}\overline{\omega}_c \cdot \overline{E}$$

Identifying $\overline{J} = Nq\overline{v}$ and $\omega_p^2 = Nq^2/m\epsilon_0$, we obtain

$$-i\omega\epsilon_o\omega_p^2\overline{E} + \omega^2\overline{J} = \epsilon_o\omega_p^2\overline{\omega}_c \times \overline{E} + \omega_c^2\overline{J} - i\epsilon_o\frac{\omega_p^2}{\omega}\overline{\omega}_c(\overline{\omega}_c \cdot \overline{E})$$

we thus have

$$\overline{J} = \frac{-i\omega\epsilon_0}{\omega^2 - \omega_c^2}\left\{i\frac{\omega_p^2}{\omega}\overline{\omega}_c \times \overline{E} - \omega_p^2\overline{E} + \frac{\omega_p^2}{\omega^2}\overline{\omega}_c\overline{\omega}_c \cdot \overline{E}\right\}$$

From Ampere's law, we find, since $\overline{\omega}_c = qB_0/m$,

$$\nabla \times$$

H = -i $\omega\epsilon_0\overline{E} + \overline{J} = i\omega\left\{i\epsilon_g \times \overline{E} + \epsilon\overline{E} + \frac{\omega_p^2\omega_c^2\epsilon_0}{\omega^2(\omega^2-\omega_c^2)} \cdot \overline{E}\right\} = -i\omega\overline{\overline{\epsilon}} \cdot \overline{E}$

where ϵ_g and ϵ are as defined. Writing $\overline{\overline{\epsilon}}$ in the matrix form, we find

$$\epsilon_z = \epsilon_0\left[1 - \frac{\omega_p^2/\omega^2}{1-\omega_c^2/\omega^2} + \frac{\omega_p^2\omega_c^2/\omega^4}{1-\omega_c^2/\omega^2}\right] = \epsilon_0\left[1 - \frac{\omega_p^2}{\omega^2}\right]$$

Carrying on the inverse of $\overline{\overline{\epsilon}}$, we find for $\overline{\overline{\kappa}} = \overline{\overline{\epsilon}}^{-1}$

$$\kappa = \frac{\epsilon}{\epsilon^2 - \epsilon_g^2} = \frac{1}{\epsilon_0}\left[\frac{1-\omega_p^2/\omega^2 - \omega_c^2/\omega^2}{(1-\omega_p^2/\omega^2)^2 - \omega_c^2/\omega^2}\right]$$

$$\kappa_g = \frac{-\epsilon_g}{\epsilon^2 - \epsilon_g^2} = \frac{1}{\epsilon_0}\left[\frac{\omega_c\omega_p^2/\omega^3}{(1-\omega_p^2/\omega^2)^2 - \omega_c^2/\omega^2}\right]$$

$$\kappa_z = \frac{1}{\epsilon_z} = \frac{1}{\epsilon_0}\left[\frac{1}{1-\omega_p^2/\omega^2}\right]$$

It is easily shown that $\overline{\overline{\epsilon}} \cdot \overline{\overline{\kappa}} = \overline{\overline{I}}$.

In the case of an infinitely strong magnetic field, $\omega_c \to \infty$ and we have

$$\epsilon = \epsilon_0 \qquad \kappa = \frac{1}{\epsilon_0}$$

$$\epsilon_g = 0 \qquad \kappa_g = 0$$

$$\epsilon_z = \epsilon_0\left(1 - \frac{\omega_p^2}{\omega^2}\right) \qquad \kappa_z = \frac{1}{\epsilon_0}\left[\frac{1}{1-\omega_p^2/\omega^2}\right]$$

and the medium becomes a uniaxial plasma.

Solution P2.4

$$\bar{E} = \hat{x}e^{ikz} = \frac{1}{2}\left[(\hat{x} + i\hat{y}) + (\hat{x} - i\hat{y})\right]e^{ikz}$$

6.630 Solution to Problem Set 4

Solution P4.1

Combine $\overline{\overline{\epsilon}}$ and $\overline{\overline{\sigma}}$ to form a new permittivity tensor

$$\overline{\overline{\epsilon_c}} = \begin{pmatrix} \epsilon + i\frac{\sigma}{\omega} & 0 & 0 \\ 0 & \epsilon + i\frac{\sigma}{\omega} & 0 \\ 0 & 0 & \epsilon_z + i\frac{\sigma_z}{\omega} \end{pmatrix}$$

This is a uniaxial medium. For a wave propagating in $\hat{x}$ direction, we need to consider two cases: (1) Oridanary wave, $k^{(o)} = \omega\sqrt{\mu_o(\epsilon + i\frac{\sigma}{\omega})}$; (2) Extraordinary wave, $k^{(e)} = \omega\sqrt{\mu_o(\epsilon_z + i\frac{\sigma_z}{\omega})}$.

The extraordinary wave is polarized in $\hat{z}$ direction while the ordinary wave is polarized in $\hat{y}$ direction. Since σ_z is much larger than σ, extraordinary wave will decay much faster. So the wave coming out of a piece of polaroid will be only $\hat{y}$ polarized.

Solution P4.2

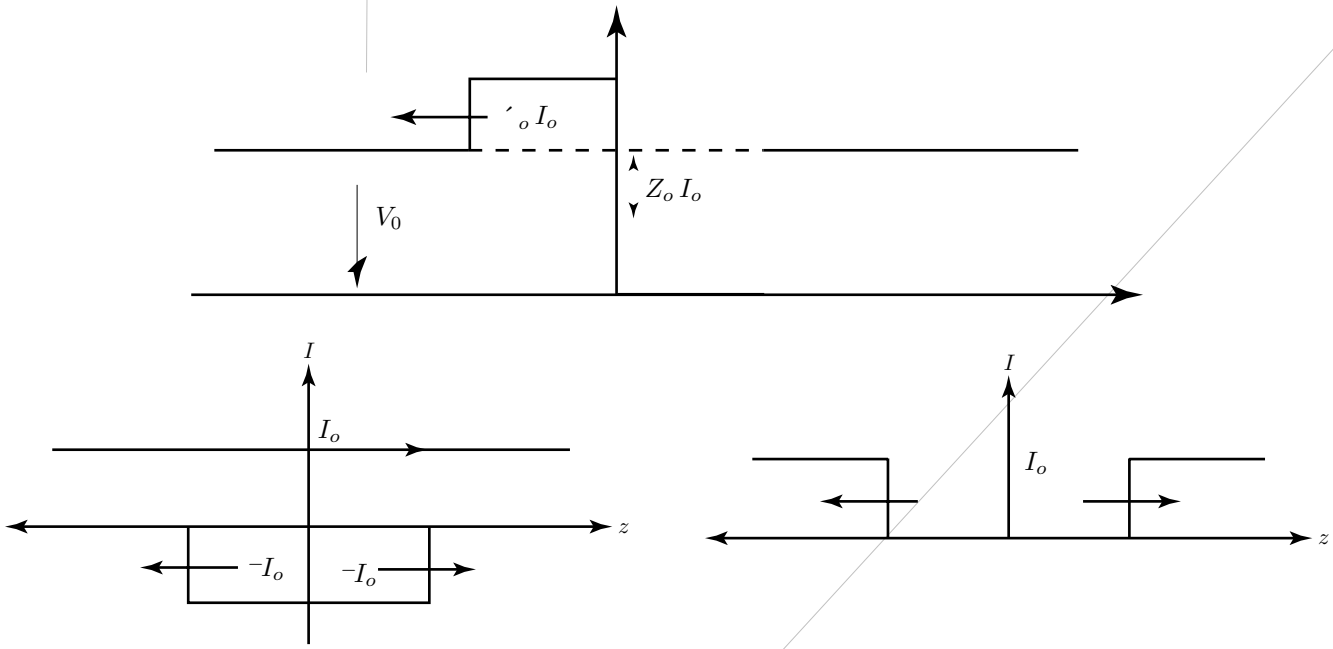
(a) $\alpha = 1$, $\beta = \pi/2$
(b) The $\hat{x}$ polarized wave is an ordinary wave. $k_y = \omega\sqrt{\mu_o 4\epsilon_o} = 2k_o$
(c) Let $(k^{(o)} - k^{(e)})d = \pi + 2n\pi$. The minimum d to satisfy this condition is $\lambda_o/2$

Solution P4.3

(a) **Method 1**

Fisrt, consider the transmission line right of the break point. When the break occurs, $I_r = I_0 + I_+ = 0$, where I_+ is the generated current. So, $I_+ = -I_0$, So, $V_+ = I_+ Z_o = -I_o Z_o$

Then, consider the transmission line left of the break point. When the break occurs, $I_l = I_0 + I_- = 0$, where I_- is the generated current. So, $I_- = -I_0$, So, $V_- = -I_- Z_o = I_o Z_o$



(a) **Method 2**

Consider the situation before the break occurs. In general there are two waves on the line, one propagating in the positive z-direction and one propagating in the negative z-direction.

$$V(z) = V_o = V_+ + V_-$$
$$I(z) = I_o = \frac{1}{Z_o}(V_+ - V_-)$$

Solving for V_+ and V_- we get,

$$V_+ = \frac{1}{2}(V_o + Z_o I_o)$$
$$V_- = \frac{1}{2}(V_o - Z_o I_o)$$

At the point of the break there is an open circuit at which the V_+ and V_- waves will be reflected generating $V'_- = \Gamma V_+$ and $V'_+ = \Gamma V_-$ respectively, where $\Gamma = 1$.

To determine the current, we can use the above analysis to relate, V'_+ to I'_+ and V'_- to I'_-.

(b) The current, I_o can be found from

$$P_{dc} = I_o V_o = 1 \times 10^9 W$$

$$\Rightarrow I_o = \frac{1 \times 10^9}{6 \times 10^5} = \frac{1}{6} \times 10^4 A$$

The peak voltage is then given by,

$$V_{peak} = V_o + Z_o I_o = 1433 \text{ kV}$$

Solution P4.4

(a) We can find the length of the transmission by,

$$\ell = \frac{(3 \times 10^8 \text{ m/s})(10^{-8} \text{ s})}{2} = 1.5\text{m}$$

(b) Since the transmission line extends to infinity, we can terminate the line with an impedance of Z_0 without introducing reflection. The terminal load, instead of R_f, becomes $Z_L = R_f || Z_0$

$$\text{Since} \quad V = V_+(1 + \Gamma_L)$$

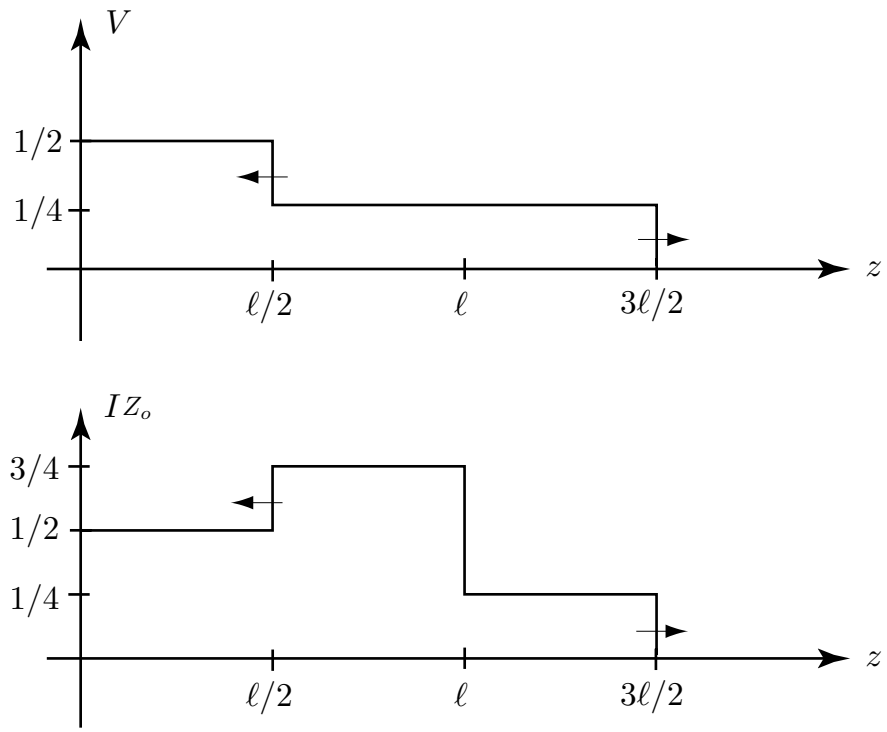
$$V = 0.25\text{V} \quad V_+ = 0.5\text{V}$$

$$\text{Hence} \quad \Gamma_L = -\frac{1}{2} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$Z_L = \frac{1}{3}Z_0, \quad \frac{R_L Z_0}{R_L + Z_0} = \frac{Z_0}{3}$$

$$R_L = \frac{Z_0}{2}$$

(c)


V
1/2
1/4
z
ℓ/2
ℓ
3ℓ/2
IZ_o
3/4
1/2
1/4
z
ℓ/2
ℓ
3ℓ/2

6.632 Solution to Problem Set 2

Solution P2.1

For ordinary waves, $< \overline{S} >$ is in the same direction with $\overline{k}$.

For extraordinary waves, the angle θ_2 between $\overline{S}$ and the optic axis is larger than θ_1 except when $\theta_1 = 0$ or $\pi/2$.

Solution P2.2

(a) We can absorb a polarization by assigning z axis to either u or v axis, since the conductivity in z is nonzero. After a sufficiently long distance, this polarization in z is completly absorbed while the other polarization still survives.

(b) This question is equivalent to calculating the penetration depth in this case. So we start with the imaginary part of k. $k_L \approx \frac{1}{2}\omega\sqrt{\mu_o\epsilon_o}\left(\frac{\sigma_z}{\omega\epsilon_z}\right) = \frac{0.1\pi}{\lambda}$. $d = 1/k_I = 3.18\lambda$.

(c) In order to avoid power absorption, the z axis must be assigned to w, and x and y can be assigned to either u and v or v and u. We can choose an incident wave as $\overline{E}_{inc} = \frac{E_o}{\sqrt{2}}(\hat{x} - \hat{y})\cos(k_o z - \omega t)$, where $x, y = u, v$. The x- and y-components of the electric field at $z = 0$ are

$$E_x = \frac{E_o}{\sqrt{2}}\cos(\omega t)\,,\quad E_y = -\frac{E_o}{\sqrt{2}}\cos(\omega t)\,.$$

Inside the slab, the x- and y-components of E field propagates with different k vectors, one being k_o while the other being $2\sqrt{3}k_o$. To get a circularly polarized wave at $z = d$, we need $(2\sqrt{3}k_o - k_o)d = \pi/2$. So $d = \frac{\lambda}{4(2\sqrt{3}-1)}$. In this case, the E field at $z = d$ is right-handed circularly polarized.

Solution P2.3

(a) From the expression of $\hat{z} \times \overline{H}_1$, we can obtain that $\overline{\overline{z}} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

(b) $\frac{d\overline{M}}{dt} = \frac{d\overline{M}_1}{dt} = -i\omega\overline{M}_1 \approx g\mu_o M_o \hat{z} \times \overline{H}_1 - g\mu_o H_o \hat{z} \times \overline{M}_1 = g\mu_o(M_o\overline{\overline{z}} \cdot \overline{H}_1 - H_o\overline{\overline{z}} \cdot \overline{M}_1)$.

From the above equation we can get the relation between $\overline{H}_1$ and $\overline{M}_1$. From the definition $\overline{B}_1 = \mu_o(\overline{H}_1 + \overline{M}_1) = \overline{\overline{\mu}} \cdot \overline{H}_1$, it can be obtained that

$$\overline{\overline{\mu}} = \begin{bmatrix} \mu & i\mu_g & 0 \\ -i\mu_g & \mu & 0 \\ 0 & 0 & \mu_z \end{bmatrix}$$

where $\mu = \frac{\omega^2 - g^2\mu_o^2 H_o^2 - g^2\mu_o^2 H_o M_o}{\omega^2 - g^2\mu_o^2 H_o^2}\mu_o$, $\mu_g = \frac{-\omega g\mu_o M_o}{\omega^2 - g^2\mu_o^2 H_o^2}\mu_o$, $\mu_z = \mu_o$.

(c) We first get

$$\overline{\overline{\nu}} = \begin{bmatrix} \nu & i\nu_g & 0 \\ -i\nu_g & \nu & 0 \\ 0 & 0 & \nu_z \end{bmatrix}$$

where $\nu = \frac{\mu}{\mu^2 - \mu_g^2}$, $\nu_g = \frac{-\mu_g}{\mu^2 - \mu_g^2}$, $\nu_z = \frac{1}{\mu_o}$. Similar to the derivations in the textbook, we eventually get the wave equation in kDB system as follows

$$\begin{bmatrix} u^2 - \kappa\nu & -i\kappa\nu_g\cos\theta \\ i\kappa\nu_g\cos\theta & u^2 - \kappa(\nu\cos^2\theta + \nu_z\sin^2\theta) \end{bmatrix} \cdot \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} = 0$$

.

So

$$\frac{D_2}{D_1} = -\frac{B_1}{B_2} = \frac{(\nu - nu_o)\sin^2\theta \pm \sqrt{(\nu-\nu_o)^2\sin^4\theta + 4\nu_g^2\cos^2\theta^2}}{2i\nu_g\cos\theta}$$

.

Define $\tan 2\psi = \frac{2\nu_g\cos\theta}{(\nu-\nu_z)\sin^2\theta}$, then $\frac{D_2}{D_1} = -i\cot\psi$ or $\frac{D_2}{D_1} = i\tan\psi$. Both are elliptically polarized.

So Faraday rotation exists.

6.632 Solution to Problem Set 6

Solution P5.1

(a) Using the notation for TE wave, $\mathrm{p}_{12} = 2$.

$$T_{12} = \frac{1}{1+p_{12}} = \frac{2}{3}$$

$$R_{12} = \frac{1-p_{12}}{1+p_{12}} = -\frac{1}{3}$$

Since the wave is incident normally, we could also use TM notation. Then, $p_{01} = 1/2$, $R_{12} = 1/3$, and $T_{12} = 4/3$.

(b)

$$< \overline{S}_i > + < \overline{S}_r >= \hat{z}\frac{1}{2}\frac{k_1}{\omega\mu_1}(1-|R|^2)$$

$$< \vec{S}_t >= \hat{z}\frac{1}{2}\frac{k_2}{\omega\mu_2}|T|^2$$

Power is conserved since we know from application of the boundary conditions $1-|R|^2 = p_{12}|T|^2$.

(c)

$$< \overline{G}_i > + < \overline{G}_r >= \hat{z}\frac{\epsilon_1}{2}\frac{k_1}{\omega}(1-|R|^2)$$

$$< \vec{G}_t >= \hat{z}\frac{\epsilon_2}{2}\frac{k_2}{\omega}|T|^2$$

Momentum is not conserved for the given parameters since $1-|R|^2 \neq 8|T|^2$.

(d) The radiation pressure of an electromagnetic wave is $p = \frac{1}{2}\epsilon|E|^2$. Therefore the pressure on the half space is

$$\bar{F} = \overline{p}_i - \overline{p}_r - \overline{p}_t = \hat{z}\frac{1}{2}\left[\epsilon_1(1+|R|^2) - \epsilon_2|T|^2\right] = -\hat{z}\frac{\epsilon_0}{3}$$

(e) The half space moves toward the incident wave.

Solution P5.2

(a) $\nabla\times\bar{H} = -i\omega\bar{D} = -i\omega(\epsilon_0\bar{E}+\bar{P})$ Therefore, we may isolate the contribution from the material as $\nabla\times\bar{H} + i\omega\epsilon_0\bar{E} = \text{-i}\,\omega\bar{P} \equiv \bar{J}_e$

(b) $\overline{J}_b = -i\omega(\epsilon_R - \epsilon_0)\overline{E}$

(c)

$$\overline{H} = \frac{1}{\omega\mu}\overline{k}\times\overline{E} = \hat{y}\frac{k_R + ik_I}{\omega\mu_0}E_0 e^{k_I z}e^{ik_R z}$$

(d) $< \overline{f}_b >= -\hat{z}\frac{1}{2}k_I(\epsilon_R-\epsilon_0)|E|^2$, where $|E|^2 = |E_0|^2 e^{2k_i z}$.

(e) $< \overline{f}_c >= \hat{z}\frac{1}{2}k_R\epsilon_I|E|^2 = -\hat{z}\frac{1}{2}\frac{n}{c}\mathrm{Re}\{\nabla\cdot\overline{S}\}$ since Poynting's theorem tells us that $\omega\epsilon_I|E|^2 = -\mathrm{Re}\{\nabla\cdot\overline{S}\}$.

Solution P5.3

(a) $\theta_c = 60^\circ$.

(b) The Brewster angle is $\theta_b = 40.9°$.

(c) It is impossible, $\sin\theta < \tan\theta$ for any θ between $0°$ and $90°$.

Solution P5.4

(a) For E layer, $f_p = 2.84\,\text{MHz}$.
For F layer, $f_p = 6.95\,\text{MHz}$.

(b) In E layer, $\theta_t = \sin^{-1}\left[1.04\sin\theta\right]$.
In F layer, $\theta_t = \sin^{-1}\left[1.39\sin\theta\right]$.

(c) For E layer total reflection happens when $f < \dfrac{2}{\sqrt{3}} f_p = 3.27\,\text{MHz}$.

For F layer total reflection happens when $f < \dfrac{2}{\sqrt{3}} f_p = 8.03\,\text{MHz}$.

6.630 Solution to Problem Set 7

Solution P7.1

$k_x = k_o \sin\theta\,,\ k_z = \sqrt{\omega^2\mu_o(\epsilon_o + i\sigma/\omega) - k_x^2} \approx \sqrt{\omega\mu_o\sigma}(\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}})$

$\theta_t = \tan^{-1}(k_x/k_{zR}) = tan^{-1}(\frac{\sqrt{2}k_x}{\sqrt{\omega\mu_o\sigma}}) \approx 0$

Solution P7.2

For a TEM wave, we can treat it as either a TE or a TM wave. Though the reflection coefficients R^{TE} and R^{TM} have different forms, they represent the same field and can be converted to each other. Therefore for normal incidence, no mather using R^{TE} or R^{TM}, we should get the same field and the same reflectivity and transmissivity.

Solution P7.3

$\overline{E}_r = (-\frac{1}{\sqrt{2}}\hat{x} + \frac{1}{\sqrt{2}}\hat{z} - i\hat{y})e^{i\frac{1}{\sqrt{2}}k_o x - i\frac{1}{\sqrt{2}}k_o z}$

The incident wave is L. H. C. P. The reflected wave is R. H. C. P.

Solution P7.4

(a) $\phi = 2\theta_2 - 2(\theta_1 - \theta_2) = 2(2\theta_2 - \theta_1)$

(b) $\frac{d\phi}{d\theta_1} = \frac{d}{d\theta_1}\left[2\sin^{-1}(\frac{\sin\theta_1}{n}) - \theta_1\right] = 0 \Rightarrow \sin\theta_1 = \sqrt{(4-n^2)/3}$
For $n = 4/3 = 1.33\,,\ \sin\theta = 0.86 \Rightarrow \phi = 2(2\theta_2 - \theta_1) \simeq 42^\circ\,.$

(c) For red light, $\theta_1 = 59.58^\circ\ \theta_2 = 40.42^\circ\,,\ \phi_{\max} = 42.52^\circ\,,$ and $\theta_s = 137.5^\circ\,;$ for violet light $\theta_1 = 58.89^\circ\,,\ \theta_2 = 39.64^\circ\,,\ \phi_{\max} = 40.78^\circ\,,$ and $\theta_s = 139^\circ\,.$ The outer arc of the rainbow will be red, and violet will be on the inner arc of the rainbow.

6.630 Solution to Problem Set 9

Solution P9.1

$k_c = \pi/d\,,\ f_c = 1.88$ kHz.

Solution P9.2

(a) $m < \frac{d\omega\sqrt{\mu_o\epsilon_o}}{\pi} = \frac{d\omega}{\pi c} = \frac{30\times10^9\times2\sqrt{3}\pi\times10^{-2}}{3\times10^8\pi} = 3.46$.
The possible guided modes are TM_m ($m = 0, 1, 2, 3$) and TE_m ($m = 1, 2, 3$).

(b) For TM_2 mode,

$$\overline{E} = \frac{H_o}{\epsilon_o\omega}\left[\hat{x}k_z\cos\frac{2\pi x}{d}e^{ik_z z} - \hat{z}\frac{2\pi i}{d}\sin\frac{2\pi x}{d}e^{ik_z z}\right]$$

(c) $v_p = \frac{\omega}{k_z} = \frac{30\times10^9}{100\sqrt{2/3}} = 3.67\times10^8(m/s)$.

$v_g = \frac{d\omega}{dk_z} = \frac{k_z}{\omega\mu_o\epsilon_o} = \frac{c^2k_z}{\omega} = \frac{c^2}{v_p} = \frac{(3\times10^8)^2}{3.67\times10^8} = 2.45\times10^8(m/s)$.

(d) Since $\epsilon_1 = 3\epsilon_o > \epsilon_0$, there is no total reflection for any modes

(e) The Brewster angle for the medium is given by $\theta_B = \tan^{-1}\sqrt{3} = 60^\circ$. Now let's see if there is some mode that is incident at 60° . We find that,

$$k_x = \frac{m\pi}{d} = 100K_o\sin 60^\circ \Rightarrow \frac{100m\pi}{\sqrt{3}} = 100(2\pi)\frac{\sqrt{3}}{2} \Rightarrow m = 3$$

So the TM_3 wave can be totally transmitted.

Solution P9.3

Since cutoff occurs when $k_z = k_o$, i.e. Goos-Hanchen shift is zero, the guidance condition now is $k_x d = m\pi = \sqrt{k_{cm}^2 - k_o^2}d$. Noting $k_o^2 = k_{cm}^2\frac{\mu_o\epsilon_o}{\mu\epsilon}$, we can get $k_{cm} = \frac{m\pi}{d\sqrt{1-\frac{\mu_o\epsilon_o}{\mu\epsilon}}}$. Let $k > k_{cm}$, we can get $m = 0, 1, 2$. The propagating modes are $\mathrm{TE}_0\,, \mathrm{TM}_0\,,\ \mathrm{TE}_1\,, \mathrm{TM}_1\,,\ \mathrm{TE}_2\,, \mathrm{TM}_2$. Their corresponding cutoff frequencies are 0, 0, 12 GHz, 12 GHz, 24 GHz, 24 GHz, respectively. Note that the cutoff frequencies for TE and TM are degenerate. Within the frequency range 0-12GHz, only TE_0 and TM_0 exist.

6.630 Solution to Problem Set 10

Solution P10.1

(a) $f_{c10} = \frac{\omega_{c10}}{2\pi} = \frac{c}{2\pi}\left(\frac{\pi}{a}\right) = \frac{3\times10^8}{2\pi} \times \frac{\pi}{6.55} = 22.9\,(MHz) < f < f_{c01} = \frac{c}{2\pi}\left(\frac{\pi}{b}\right) = \frac{3\times10^8}{2\pi} \times \frac{\pi}{4.19} = 35.8\,(MHz)$

(b) An AM radio operates in the range of 500 to 1600 (KHz) is below the cutoff frequency of the fundamental mode TE_{10}. Therefore AM signals can not be received in the tunnel.

(c) FM singals operate in the range of 88 to 108 (MHz) can be received in the tunnel.

Solution P10.2

In the region of $z > 0$,

$$H_y = \Sigma_{m=1}^{\infty} H_m^{(1)} \cos(m\pi x/d) e^{ik_z z},$$

$$E_x = \Sigma_{m=1}^{\infty} \frac{k_z H_m^{(1)}}{\omega\epsilon} \cos(m\pi x/d) e^{ik_z z}.$$

In the region of $z < 0$,

$$H_y = \Sigma_{m=1}^{\infty} H_m^{(2)} \cos(m\pi x/d) e^{-ik_z z},$$

$$E_x = \Sigma_{m=1}^{\infty} \frac{-k_z H_m^{(1)}}{\omega\epsilon} \cos(m\pi x/d) e^{-ik_z z}.$$

At the boundary $z = 0$,

$$\Sigma_{m=1}^{\infty}(H_m^{(2)} - H_m^{(1)}) \cos\frac{m\pi x}{d} = J_s \cos\frac{3\pi x}{d},$$

$$\Sigma_{m=1}^{\infty} \frac{k_z H_m^{(1)}}{\omega\epsilon} \cos\frac{m\pi x}{d} = \Sigma_{m=1}^{\infty} \frac{-k_z H_m^{(2)}}{\omega\epsilon} \cos\frac{m\pi x}{d}.$$

So $m = 3$ and $H_m^{(2)} = -H_m^{(1)} = J_s/2$.
In the regin of $z > 0$, $H_y = -\frac{J_s}{2}\cos\frac{3\pi x}{d} e^{ik_z z}$, $E_x = -\frac{k_z J_s}{2\omega\epsilon}\cos\frac{3\pi x}{d} e^{ik_z z}$ and $E_z = \frac{iJ_s}{2\omega\epsilon}(\frac{3\pi}{d})\sin\frac{3\pi x}{d} e^{ik_z z}$.
In the regin of $z < 0$, $H_y = \frac{J_s}{2}\cos\frac{3\pi x}{d} e^{-ik_z z}$, $E_x = -\frac{k_z J_s}{2\omega\epsilon}\cos\frac{3\pi x}{d} e^{-ik_z z}$ and $E_z = \frac{-iJ_s}{2\omega\epsilon}(\frac{3\pi}{d})\sin\frac{3\pi x}{d} e^{-ik_z z}$.

Solution P10.3

(a) Only TE_{10} mode is propagating. $t = \ell/v_g = 1.8\ \mu\mathrm{s}$.

(b) Three modes, TE_{10}, TE_{01} and TE_{20}, whose cutoff frequencies are 5 GHz, 10GHz, and 10GHz, respectively.
For TE_{10}, $t = 1.14\ \mu\mathrm{s}$. For TE_{01} and TE_{20}, $t = 3.28\ \mu\mathrm{s}$.

6.630 Solution to Problem Set 8

Solution P8.1

$\theta_c = \arcsin^{-1}(n_2/n_1) = 89.7^\circ$

Solution P8.2

(a) The Brewster angle for $\epsilon_t = 9$ is $\theta_B = \tan^{-1}\sqrt{\epsilon_t} = \tan^{-1}\sqrt{9} = 71.57^\circ$.
(b) The dominant portion of the sun glares is TE polarized wave. The polaroid glasses absorb the TE component of the incident light, thus the TM component reaches the eyes after passing through the polaroid glasses.

Solution P8.3

The critical angle $\theta_c = \sin^{-1}\sqrt{\epsilon}$

$$\overline{E}_i = \hat{y}\frac{2E_0}{3\sqrt{2}}e^{ik_x x + ik_z z} + E_0\left[\frac{\hat{z}}{2\sqrt{2}} - \frac{\hat{x}}{2\sqrt{6}}\right]e^{ik_x x + ik_z z - i\pi/2}$$

$$\overline{E}_r = \hat{y}\frac{E_0}{3\sqrt{2}}e^{-ik_x x + ik_z z}$$

(c.i) $\phi = 0$
(c.ii) constant, maximum value $2E_0$

6.632 Solution to Problem Set 5

Solution P5.1

At the cutoff, the Goos-Hanchen shift at $x = d$ is zero. Combing guidance condition and dispersion relations, we can get

$$k_{cm} = \frac{m\pi + \tan^{-1}\frac{\epsilon_1\sqrt{\epsilon_2-\epsilon_0}}{\epsilon_0\sqrt{\epsilon_1-\epsilon_2}}}{d\sqrt{1-\epsilon_2/\epsilon_1}}$$

Solution P5.2

(a) $\overline{J}_s = \hat{z}\frac{2k_0}{\omega\mu}e^{i\sqrt{3}k_0 z/2}$

(b) The waves radiated by the surface current sheet are

$$\overline{E}_+ = (\hat{x}\sqrt{3}/2 - \hat{z}1/2)e^{ik_0 x/2}e^{ik_0 z\sqrt{3}/2}$$

$$\overline{E}_- = (-\hat{x}\sqrt{3}/2 - \hat{z}1/2)e^{-ik_0 x/2}e^{ik_0 z\sqrt{3}/2}$$

which cancel the incident wave in region $x > 0$ and form the reflected wave in region $x < 0$.

Solution P5.3

(a)

$$\frac{d^2S_\rho}{d\ell d\lambda} = \frac{d^2S_\rho}{d\ell d\omega}\frac{d\omega}{d\lambda} = \frac{\mu q^2}{4\pi}\omega(1-\frac{1}{n^2\beta^2})\frac{2\pi c}{\lambda^2}$$

$$\frac{d^2N}{d\ell d\lambda} = \frac{q^2 c}{2\lambda^2\hbar}\mu(1-\frac{1}{n^2\beta^2})$$

Since $1-\frac{1}{n^2\beta^2} = \sin^2\theta$, $\frac{dN}{d\ell} \propto \frac{d\lambda}{\lambda^2}\sin^2\theta$.

(b) $\theta = 3.6^\circ$

(c) $\ell = 0.528$ m